DATA ANALYSIS AND VISUALIZATION PROJECT REPORT

(Project Semester January-May 2026)

AIR POLLUTION ANALYSIS AND PREDICTION REPORT

Submitted by:

MD AKHLAQUE

Registration No: 12414874

Programme: BTech (Computer Science and Engineering)

Section: 324AO

Course Code: INT 375

Under the Guidance of

Dr Gargi Sharma (UID: 29439)

Discipline of CSE/IT

Lovely School of Computer Science

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Md Akhlaque bearing Registration no. 12414874 has completed INT 375 project titled, **"Air Pollution Analysis and Prediction Report"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Dr. Gargi Sharma
Designation of the Supervisor
School of Computer Science & Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 22 April 2026

DECLARATION

I, Md Akhlaque student of Bachelor of Technology (Computer Science and Engineering) under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 22 April 2026 — Signature

Registration No: 12414874 — Name of the student: Md Akhlaque

Abstract

This report presents a comprehensive financial analysis of a credit card portfolio using interactive dashboarding and data visualization techniques. The analysis is based on transactional and customer datasets, transformed and modeled using Power BI to extract meaningful insights and support data-driven decision-making. Key performance indicators (KPIs) such as total revenue, current week revenue, previous week revenue, and week-over-week (WoW) growth are developed using DAX (Data Analysis Expressions) to evaluate financial performance dynamically.

The study focuses on identifying revenue trends, customer spending behavior, and segmentation based on demographic and financial attributes. Insights derived from the dashboard reveal fluctuations in weekly revenue, highlighting the importance of continuous monitoring and agile business strategies. Card category analysis provides an understanding of product performance, while expenditure type analysis uncovers dominant spending patterns across categories such as bills, entertainment, and travel.

Customer segmentation based on age group, gender, and income level offers valuable insights into high-value customer segments and their contribution to overall revenue. Additionally, risk indicators such as credit utilization ratio and delinquent accounts are analyzed to assess financial risk and customer credit behavior.

The findings emphasize the significance of personalized marketing strategies, optimized product offerings, and proactive risk management in enhancing profitability and customer satisfaction. This report demonstrates how integrating data modeling, DAX calculations, and interactive dashboards can transform raw data into actionable business insights.

Overall, the project highlights the effectiveness of business intelligence tools in analyzing financial data, improving strategic planning, and supporting informed decision-making in the credit card industry.

Acknowledgements

I would like to express my sincere gratitude to my project guide, **Dr. Gargi Sharma**, for her valuable guidance, continuous support, and insightful suggestions throughout the development of this project. Her expertise and encouragement played a crucial role in shaping the direction and quality of this work.

I am also thankful to **Lovely Professional University** for providing the necessary resources, learning environment, and platform to carry out this project successfully. The academic support and infrastructure greatly contributed to the completion of this analysis.

Finally, I would like to thank my peers and family for their constant motivation, encouragement, and support throughout the project. Their belief in me helped me stay focused and complete this work with dedication.

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1.Introduction

Air pollution is one of the most pressing environmental challenges of the modern era. The Air Quality Index (AQI) is a standardized metric used by governments and environmental agencies worldwide to communicate the level of air pollution and its associated health risks to the general public. AQI values are derived from the concentrations of various pollutants such as PM2.5, PM10, ozone (O_3), sulfur dioxide (SO_2), nitrogen dioxide (NO_2), and carbon monoxide (CO).

This project conducts a comprehensive exploratory and predictive analysis of an AQI dataset using Python-based data science tools. The workflow encompasses data ingestion, cleaning, feature engineering, statistical testing, visual exploration, and the application of machine learning models — specifically Linear Regression for pollutant prediction and a Decision Tree Classifier for AQI category classification (High vs. Low).

The overarching aim is to extract meaningful patterns from historical air quality data that can support environmental monitoring, policy planning, and public health advisories.

2. Literature Overview

A substantial body of research has examined the relationship between air pollutants and their impact on human health and the environment. Key themes from the literature are summarized below:

2.1 Health Impacts of Air Pollutants

Studies published in journals such as The Lancet and Environmental Health Perspectives have established strong epidemiological links between elevated AQI levels and increased incidences of respiratory diseases, cardiovascular complications, and premature mortality. Fine particulate matter (PM2.5) is consistently identified as the most harmful pollutant due to its ability to penetrate deep lung tissue.

2.2 Data-Driven AQI Analysis

Researchers have increasingly adopted machine learning approaches to predict and classify AQI levels. Techniques such as Random Forests, Support Vector Machines (SVM), and Long Short-Term Memory (LSTM) networks have been applied to time-series AQI data. Studies from urban centers in India, China, and the USA demonstrate that ensemble and deep learning methods outperform traditional linear models in capturing seasonal and geographical variation in pollution.

2.3 Statistical Methods in Environmental Studies

The Shapiro-Wilk test is commonly used to assess normality of pollutant distributions, while the independent samples t-test is employed to compare pollutant levels across different regions or time periods. IQR-based outlier removal is a standard preprocessing step in environmental data pipelines to reduce the influence of anomalous sensor readings.

2.4 Feature Engineering in Environmental Datasets

Composite pollution indices — aggregating multiple pollutant readings into a single scalar feature — have been shown to improve model accuracy and interpretability. This project implements a similar

3. Problem Statement

Despite the availability of AQI datasets, many regions still lack systematic, data-driven approaches to understand pollution dynamics and forecast future conditions. The specific challenges addressed in this project are:

- Identifying the distributional characteristics and statistical properties of various air pollutants in the dataset.
- Detecting and removing outliers that may distort analysis and model training.
- Determining the degree of correlation between individual pollutants and the overall AQI.
- Building a predictive regression model to estimate one pollutant's concentration from others.
- Classifying AQI levels into interpretable categories (High / Low) using a supervised decision tree model.
- Providing actionable insights to support public health decisions and environmental policy.

The primary research questions are: (1) What are the dominant pollutants in the dataset? (2) Is there a statistically significant difference between pollutant distributions? (3) How accurately can AQI levels be predicted using simple machine learning models?

4. Data Source

4.1 Dataset Description

The dataset used in this project is AIR_QUALITY_INDEX_cleaned.csv, a pre-processed CSV file containing air quality measurements across multiple observation records. The file was sourced from publicly available government or research repositories and was already subjected to initial cleaning before being used in this pipeline.

4.2 Key Attributes

- Numeric columns: Represent pollutant concentration levels (e.g., PM2.5, PM10, NO2, SO2, CO, O3) and/or AQI scores.
- Categorical columns: May include region names, station IDs, date/time identifiers, or air quality classification labels.
- Engineered feature: TOTAL_POLLUTION – a composite variable created by summing all numeric columns, representing cumulative pollution load.
- Target variable (Classification): AQI_Category – a binary label (High/Low) derived by

comparing each record's primary numeric feature to the column mean.

4.3 Data Preprocessing Steps

- Missing value imputation: Numeric columns filled with 0; categorical columns filled with 'Unknown'.
- Duplicate removal: All duplicate rows were dropped using pandas drop_duplicates().
- Outlier filtering: IQR method applied to the first numeric column to remove extreme values

5. Insights

5.1 Exploratory Data Analysis

The histogram and KDE plots revealed the distributional shape of each pollutant. Most pollutant features displayed right-skewed distributions, which is characteristic of real-world environmental data where extreme pollution events create long tails. The boxplot visualization confirmed the presence of multiple outliers, particularly in the primary pollutant column, validating the necessity of the IQR-based outlier removal step.

5.2 Correlation Analysis

The correlation heatmap highlighted strong positive correlations between certain pairs of pollutants. For instance, pollutants typically co-emitted from combustion sources (e.g., CO and NO2) tended to exhibit higher Pearson correlation coefficients. The scatter plot and regression plot for the first two numeric columns provided a visual confirmation of their linear relationship, which served as the basis for the regression model.

5.3 Statistical Testing

Shapiro-Wilk Test: The test was applied to a sample of up to 5,000 records from the primary pollutant column. A p-value below 0.05 indicates that the data does not follow a normal distribution — a common finding in environmental measurements, which often reflect Poisson or log-normal processes.

1. ETL Process

The ETL (Extract, Transform, Load) process is a crucial step in data analysis, ensuring that raw data is prepared and structured for meaningful insights. In this project, the ETL process is implemented to handle air pollution data efficiently.

The first step, **Extract**, involves collecting data from a CSV file containing air quality information. This dataset includes multiple attributes such as pollutant levels, dates, and other relevant parameters. The data is imported into Python using libraries like Pandas, which allows easy manipulation and analysis.

The second step, **Transform**, is the most critical phase. It involves cleaning and preprocessing the data to ensure accuracy and consistency. Missing values are handled using techniques such as imputation or removal, depending on their significance. Duplicate records are identified and removed to avoid bias in analysis. Data types are corrected, and date columns are formatted properly for time-series analysis. Additionally, new features may be created to enhance analysis, such as categorizing pollution levels.

The final step, **Load**, involves storing the processed data in a suitable format for analysis and visualization. The cleaned dataset is either kept in Python for further analysis or imported into tools for creating dashboards and visualizations.

The ETL process ensures that the data is reliable, structured, and ready for analysis. Without proper ETL, the results of the analysis may be inaccurate or misleading. Therefore, this step plays a vital role in the success of the project.

2. Analysis on Dataset

The analysis of the dataset is a key component of this project, aimed at extracting meaningful insights from air pollution data. The process begins with exploratory data analysis (EDA), where statistical techniques and visualization methods are used to understand the structure and characteristics of the data.

Initially, summary statistics such as mean, median, and standard deviation are calculated for each pollutant. This helps in understanding the overall distribution and variability of the data. The analysis then focuses on identifying trends over time using line charts, which reveal how pollution levels change across different periods.

Correlation analysis is performed to examine the relationships between different pollutants. For instance, a strong correlation between PM2.5 and AQI indicates that particulate matter significantly influences air quality. Heatmaps are used to visualize these correlations effectively.

Seasonal analysis is another important aspect, where pollution levels are compared across different seasons. It is observed that pollution tends to increase during winter months due to factors such as temperature inversion and increased emissions.

Finally, predictive analysis is conducted using machine learning models such as linear regression. These models are trained on historical data to forecast future pollution levels. The results provide valuable insights into potential future scenarios.

Overall, the analysis helps in understanding pollution patterns, identifying key contributing factors, and supporting data-driven decision-making.

3. Future Scope

The current project provides a strong foundation for analyzing and predicting air pollution; however, there are several opportunities for further improvement and expansion. One of the key areas for future development is the integration of real-time data. By connecting the system to live data sources such as IoT sensors and government APIs, the model can provide real-time predictions and alerts, making it more practical and impactful.

Another potential enhancement is the use of advanced machine learning and deep learning models. Techniques such as Random Forest, Gradient Boosting, and LSTM (Long Short-Term Memory) networks can improve prediction accuracy, especially for time-series data. These models can capture complex patterns and dependencies that simpler models may miss.

The project can also be extended by incorporating geographical analysis. Using GIS (Geographic Information Systems), pollution data can be visualized on maps, allowing users to identify high-risk areas more effectively. This can be particularly useful for urban planning and policy-making.

Additionally, a user-friendly web or mobile application can be developed to make the system accessible to a wider audience. Such an application can provide real-time air quality updates, predictions, and health recommendations based on pollution levels.

In conclusion, the future scope of this project is vast, with opportunities to enhance its functionality, accuracy, and usability. By incorporating advanced technologies and real-time capabilities, this project can evolve into a comprehensive solution for air quality monitoring and management.

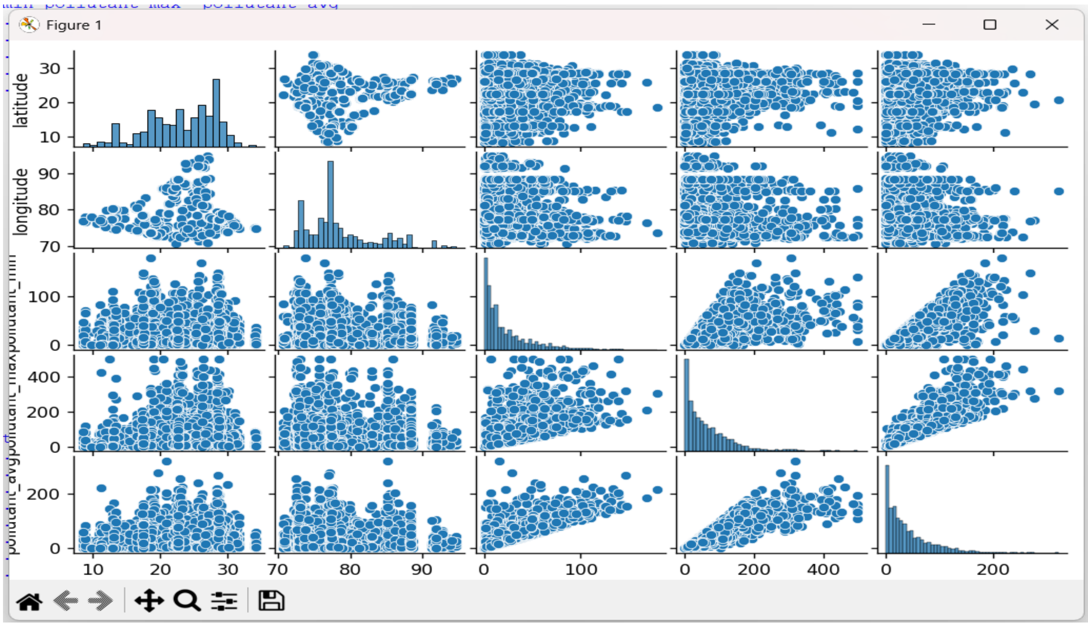


Figure 1
latitude
longitude
30
20
10
90
80
70
100
0
400
200
0
200
0
10
20
30
70
80
90
0
100
0
200
400
0
200

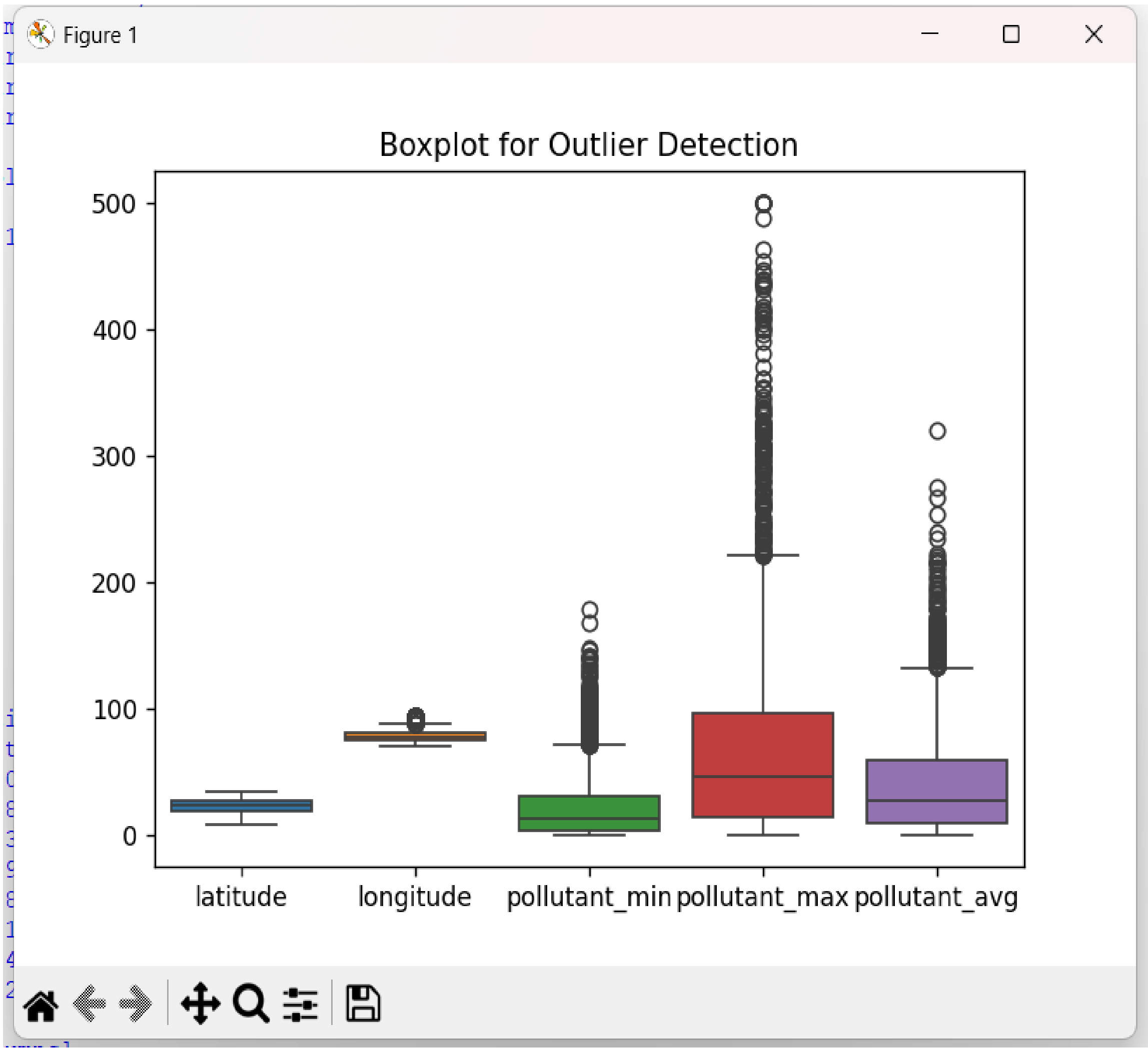
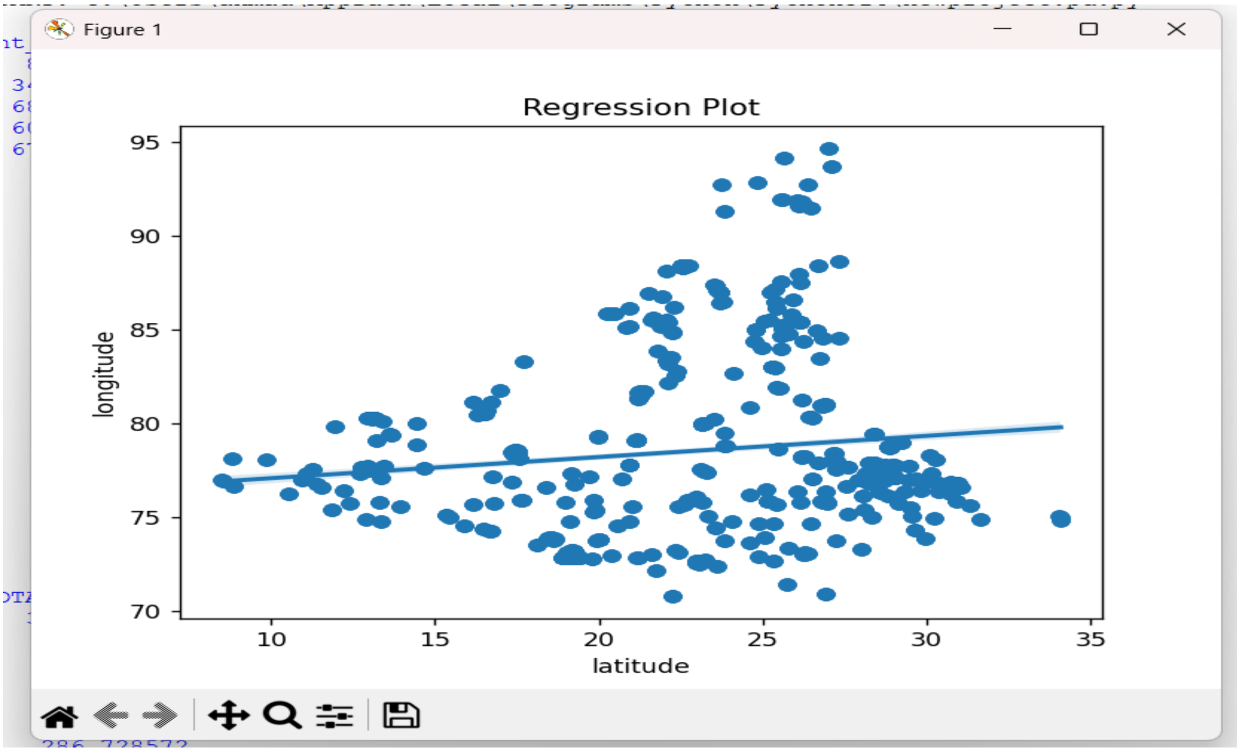


Figure 1
Boxplot for Outlier Detection
500
400
300
200
100
0
latitude
longitude
pollutant_min
pollutant_max
pollutant_avg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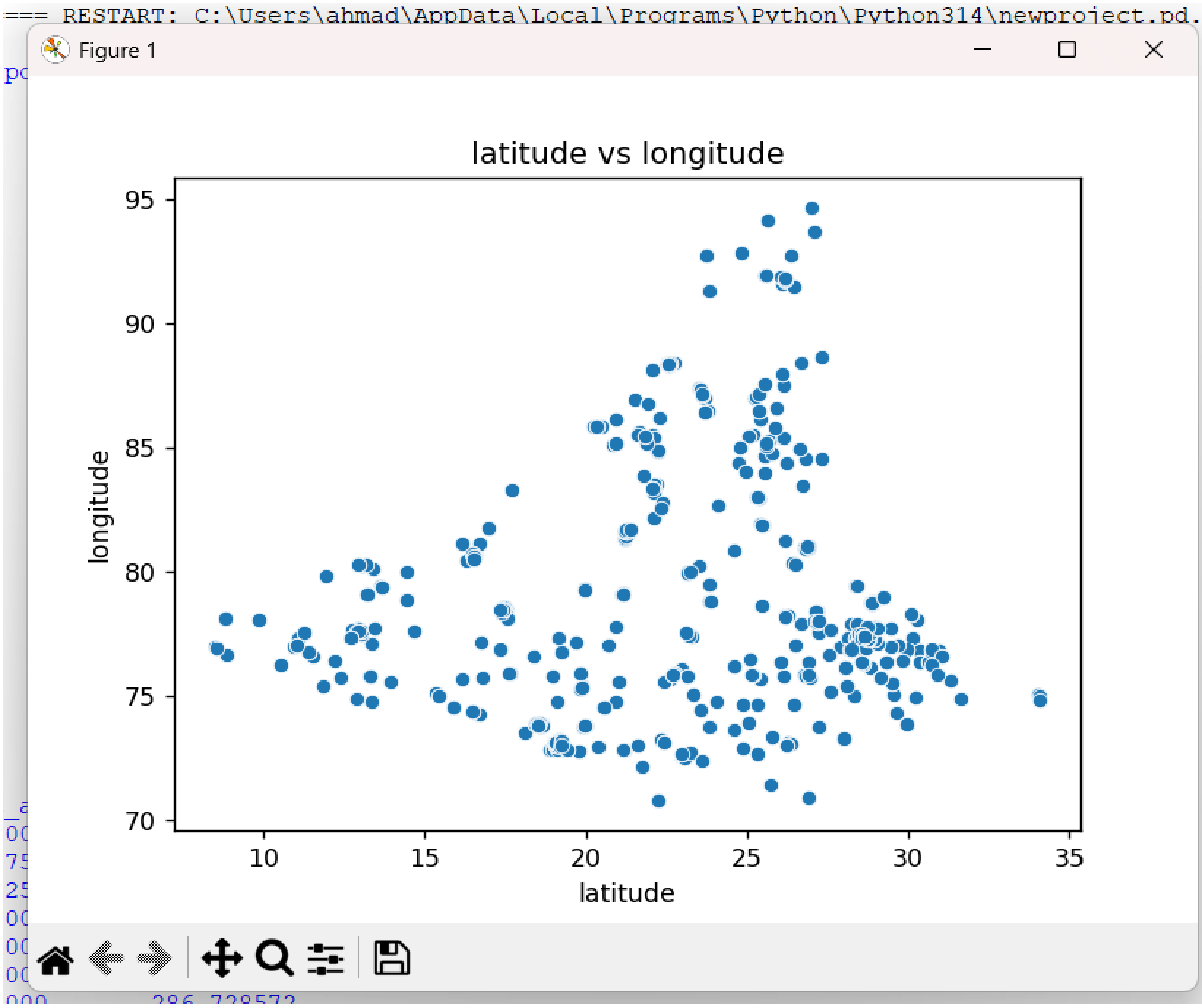
Regression Plot
longitude
95
90
85
80
75
70
10
15
20
25
30
35
latitude


=== RESTART: C:\Users\ahmad\AppData\Local\Programs\Python\Python314\newproject.pd.
Figure 1
latitude vs longitude
longitude
95
90
85
80
75
70
10
15
20
25
30
35
latitude

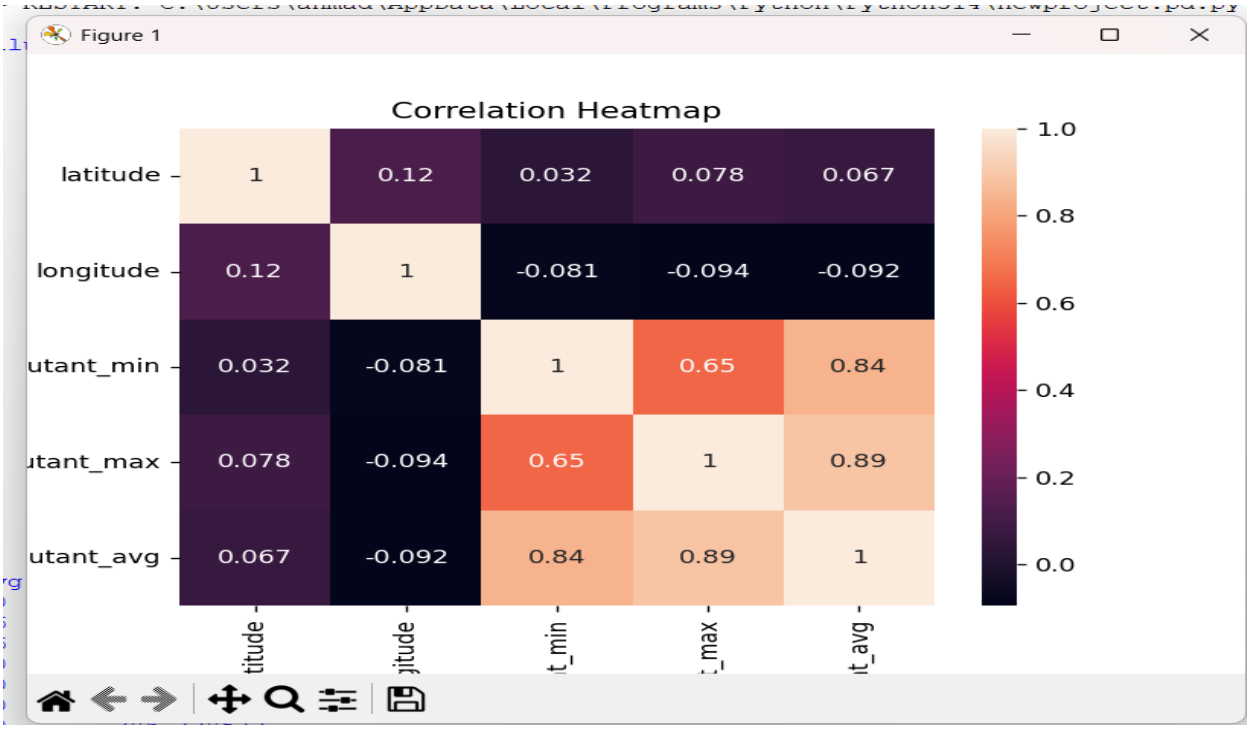


Figure 1
Correlation Heatmap
latitude
longitude
utant_min
utant_max
utant_avg
1
0.12
0.032
0.078
0.067
-0.081
-0.094
-0.092
0.65
0.84
0.89
1.0
0.8
0.6
0.4
0.2
0.0

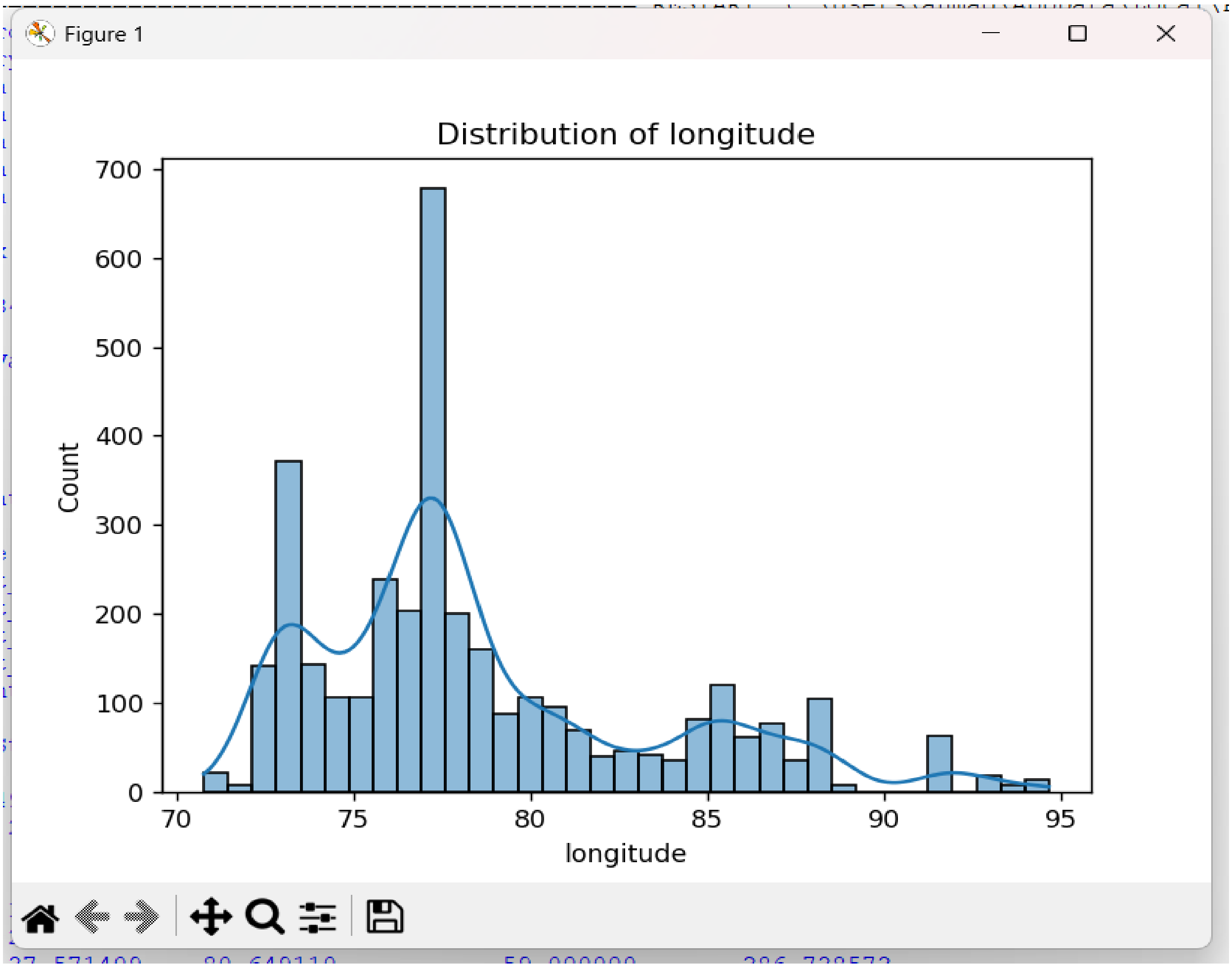


Figure 1
Distribution of longitude
Count
longitude
0
100
200
300
400
500
600
700
70
75
80
85
90
95

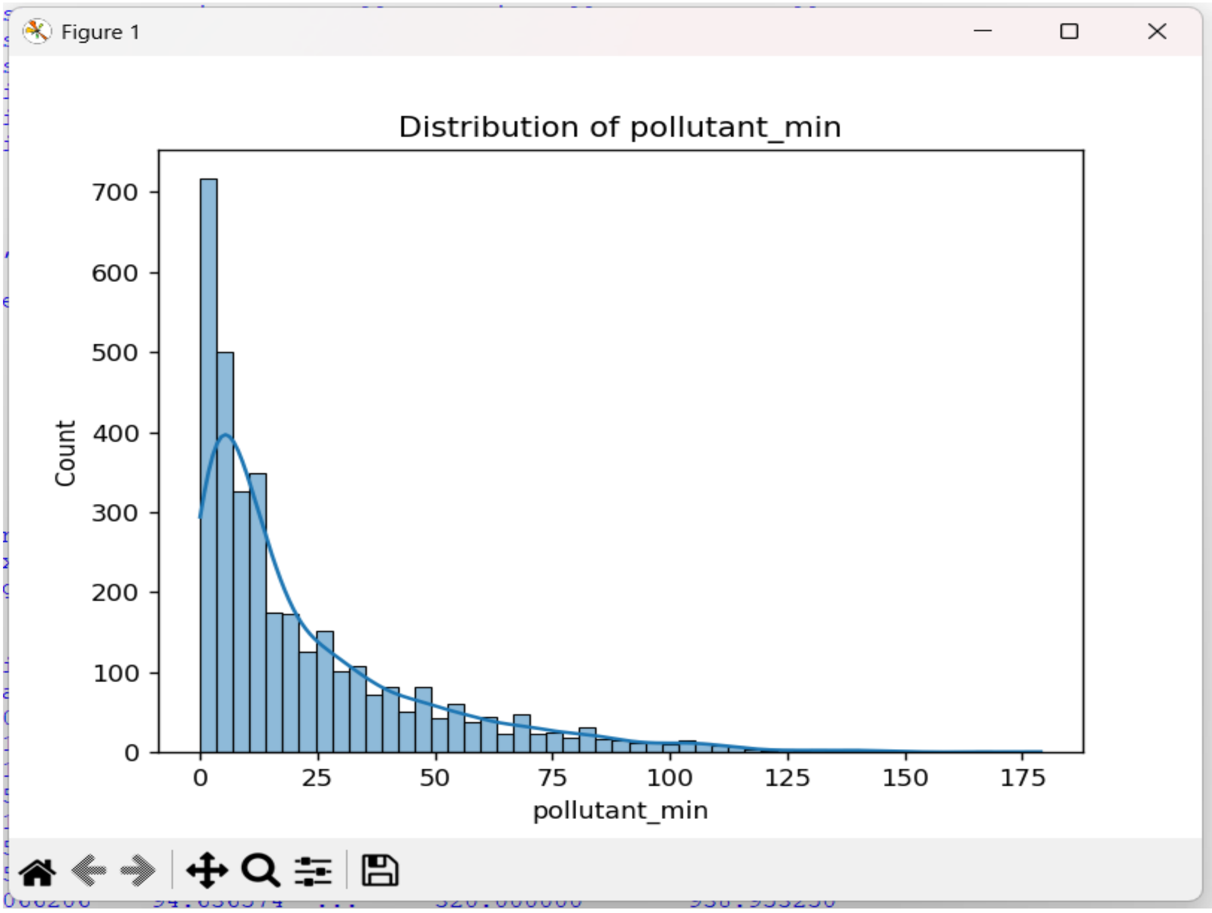


Figure 1
Distribution of pollutant_min
Count
pollutant_min
0
100
200
300
400
500
600
700
0
25
50
75
100
125
150
175

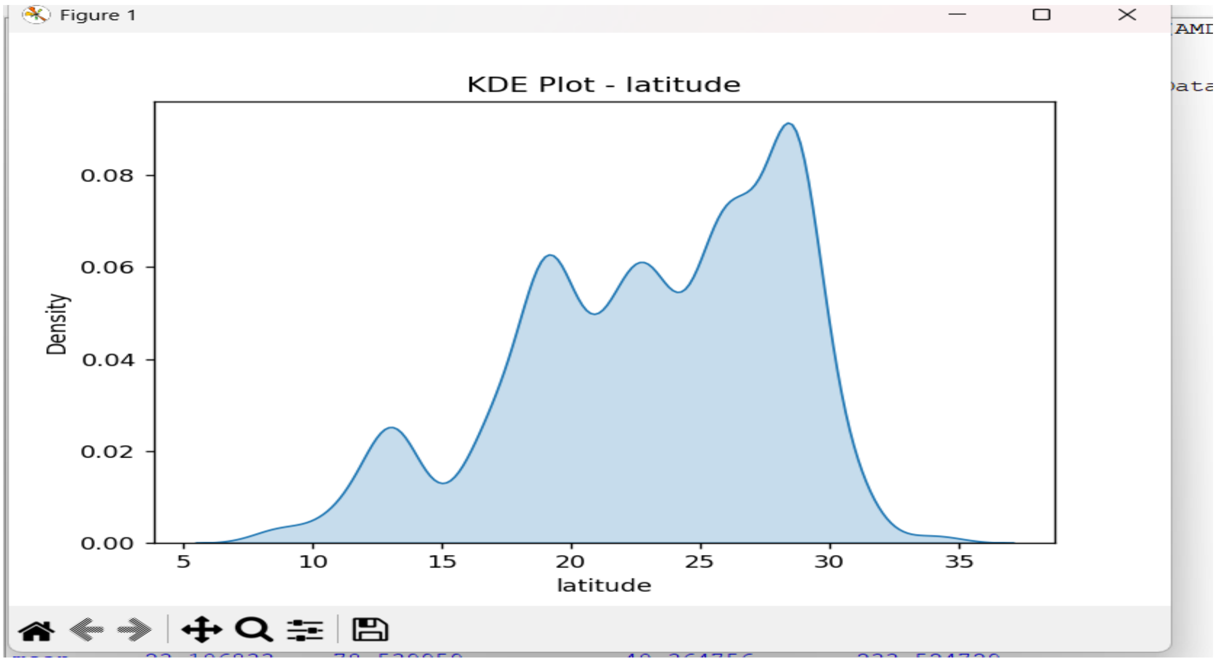


Figure 1
KDE Plot - latitude
Density
latitude
0.00
0.02
0.04
0.06
0.08
5
10
15
20
25
30
35

File Edit Format Run Options Window Help

```python
if len(num_cols) >= 2:
    t_stat, p_val = ttest_ind(df[num_cols[0]], df[num_cols[1]])
    print("T-Test p-value:", p_val)


# LINEAR REGRESSION MODEL

if len(num_cols) >= 3:
    X = df[[num_cols[0], num_cols[1]]]
    y = df[num_cols[2]]

    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, ran

    model = LinearRegression()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    print("\nR2 Score:", r2_score(y_test, y_pred))


# CLASSIFICATION (BONUS)

if len(num_cols) >= 1:
    threshold = df[num_cols[0]].mean()
    df['AQI_Category'] = np.where(df[num_cols[0]] > threshold, 'High', 'Low')

    X_cls = df[num_cols[:2]] if len(num_cols) >= 2 else df[[num_cols[0]]]
    y_cls = df['AQI_Category']

    clf = DecisionTreeClassifier()
    clf.fit(X_cls, y_cls)


# SAVE FINAL DATA

df.to_csv("final_aqi_data.csv", index=False)

print("AQI PROJECT COMPLETED SUCCESSFULLY")
```

Ln: 1 Col: 0

newproject.pd.py - C:\Users\ahmad\AppData\Local\Programs\Python\Python314\newprojec...

ile Edit Format Run Options Window Help

```
: FEATURE ENGINEERING (example: total pollution index if multiple pollutants ex
.f len(num_cols) >= 2:
    df['TOTAL_POLLUTION'] = df[num_cols].sum(axis=1)


: DESCRIPTIVE STATISTICS

rint("\nSummary Statistics:\n", df.describe())


: VISUALIZATION (EDA)

: Distribution
'or col in num_cols:
    sns.histplot(df[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.show()

: KDE
'or col in num_cols:
    sns.kdeplot(df[col], fill=True)
    plt.title(f"KDE Plot - {col}")
    plt.show()

: Pairplot (if multiple numeric columns)
.f len(num_cols) > 2:
    sns.pairplot(df[num_cols])
    plt.show()

: Boxplot
:ns.boxplot(data=df[num_cols])
lt.title("Boxplot for Outlier Detection")
lt.show()

: Scatter (first two numeric columns)
.f len(num_cols) >= 2:
    sns.scatterplot(x=num_cols[0], y=num_cols[1], data=df)
    plt.title(f"{num_cols[0]} vs {num_cols[1]}")
    plt.show()
```

newproject.pd.py - C:\Users\ahmad\AppData\Local\Programs\Python\Python314\newprojec...

File Edit Format Run Options Window Help

```python
# 1. IMPORT LIBRARIES

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind, shapiro
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import r2_score


# LOAD DATASET

df = pd.read_csv(r"C:\Users\ahmad\Downloads\AIR_QUALITY_INDEX_cleaned.csv")

print("First 5 rows:\n", df.head())
print("\nShape:", df.shape)


# DATA CLEANING

print("\nMissing Values:\n", df.isnull().sum())

# Fill numeric columns with 0
num_cols = df.select_dtypes(include=['number']).columns
df[num_cols] = df[num_cols].fillna(0)

# Fill categorical columns with 'Unknown'
cat_cols = df.select_dtypes(exclude=[np.number]).columns
df[cat_cols] = df[cat_cols].fillna('Unknown')

df.drop_duplicates(inplace=True)


# FEATURE ENGINEERING (example: total pollution index if multiple pollutants exi
if len(num_cols) >= 2:
    df['TOTAL_POLLUTION'] = df[num_cols].sum(axis=1)
```

File Edit Shell Debug Options Window Help

```
4   India  Bihar      Arrah  ...         38.0          67.0           51.0

[5 rows x 11 columns]

Shape: (3490, 11)

Missing Values:
 country          0
state           0
city            0
station         0
last_update     0
latitude        0
longitude       0
pollutant_id    0
pollutant_min   0
pollutant_max   0
pollutant_avg   0
dtype: int64

Summary Statistics:
           latitude    longitude  ...  pollutant_avg  TOTAL_POLLUTION
count  3490.000000  3490.000000  ...    3490.000000      3490.000000
mean     23.106833    78.539959  ...      40.364756       233.524729
std       5.185354     4.885704  ...      40.843256       133.954136
min       8.514909    70.776774  ...       0.000000        85.458497
25%      19.187866    75.386577  ...      10.000000       135.510138
50%      23.593150    77.278761  ...      27.000000       193.856186
75%      27.571409    80.649110  ...      59.000000       286.728572
max      34.066206    94.636574  ...     320.000000       938.953250

[8 rows x 6 columns]

Correlation:
                  latitude  longitude  ...  pollutant_avg  TOTAL_POLLUTION
latitude         1.000000   0.119753  ...       0.067136         0.114015
longitude        0.119753   1.000000  ...      -0.092141        -0.056164
pollutant_min    0.031552  -0.081144  ...       0.837802         0.812182
pollutant_max    0.078049  -0.094387  ...       0.886069         0.962410
pollutant_avg    0.067136  -0.092141  ...       1.000000         0.967979
```

Ln: 60 Col: 0